The Casimir Scale in FFGFT

A Brief Summary for the IPI Correspondence

Johann Pascher

ORCID: 0009-0000-6518-4064

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Abstract

The Fundamental Fractal Geometric Field Theory (FFGFT) makes a quantitative, parameter-free prediction for the characteristic Casimir vacuum scale: $L_\xi \approx 100{,}24\,\mu$m. The prediction is derived from the cosmic microwave background (CMB) energy density and a single dimensionless constant $\xi = 4/30{,}000$. This note summarises the derivation, the consistency with the standard Casimir formula, and the embedding of L_ξ in a five-step ξ-power ladder linking the Planck length to the Hubble length. Source documents in the FFGFT corpus: Doc 091 (Casimir-CMB), Doc 165 (Hubble), Doc 166 (synthesis).

1 Setting

FFGFT operates with a single dimensionless parameter,

$$\xi = \frac{4}{30{,}000} = 1{,}333 \times 10^{-4},$$

and the scale-anchor relation $\tilde{T} \cdot m = 1$. From these and standard physical constants, FFGFT derives a hierarchy of physical length scales without further input parameters. Two of these scales are directly relevant for the Casimir effect: a sub-Planck saturation length $L_0 = \xi \cdot \ell_P$ and a vacuum-emergent length L_ξ linked to the CMB.

2 The CMB-Casimir Relation

The central relation in Doc 091 reads:

$$\rho_{\text{CMB}} = \frac{\xi\,\hbar c}{L_\xi^4}\,. \tag{1}$$

With the measured CMB energy density $\rho_{\text{CMB}} = 4{,}17 \times 10^{-14}$ J/m^3 and $\xi = 4/30{,}000$, the relation fixes

$$L_\xi = \left(\frac{\xi\,\hbar c}{\rho_{\text{CMB}}}\right)^{1/4} \approx 100{,}24\,\mu\text{m}. \tag{2}$$

Equivalently, using the Stefan-Boltzmann form $\rho_{\text{CMB}} = (\pi^2/15)(k_B T_{\text{CMB}})^4/(\hbar c)^3$:

$$L_\xi = \left(\frac{15\,\xi}{\pi^2}\right)^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}} = 100{,}24\,\mu\text{m}\,, \tag{3}$$

which agrees with the direct evaluation to 0.11%.

The corresponding characteristic frequency is

$$f_\xi = \frac{c}{L_\xi} \approx 2{,}99\,\text{THz},$$

placing L_ξ in the same logarithmic regime as the Casimir target frequencies discussed in the IPI correspondence (Moseley's $\nu_M \approx 8{,}23$ THz / $\lambda_M \approx 36{,}4\,\mu$m), with a separation of factor $\sim 2{,}7$.

3 Consistency with the Standard Casimir Formula

The modified Casimir energy density in FFGFT is

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{240\,\xi}\,\rho_{\text{CMB}}\left(\frac{L_\xi}{d}\right)^4 . \tag{4}$$

Substituting Eq. (1) into Eq. (4) yields

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{240\,\xi} \cdot \frac{\xi\,\hbar c}{L_\xi^4} \cdot \frac{L_\xi^4}{d^4} = \frac{\pi^2 \hbar c}{240\,d^4}\,, \tag{5}$$

exactly the standard Casimir formula. The parameter ξ cancels at the operational level. The FFGFT formulation is therefore consistent with all current Casimir measurements that confirm the standard $1/d^4$ scaling, while reframing the absolute energy scale through L_ξ and ρ_{CMB}.

4 The Length-Scale Hierarchy

FFGFT places L_ξ on a five-step ladder of physical length scales, all expressible as $\xi^n \cdot \ell_P$:

Scale	**Meaning**	**Length [m]**	$\xi^n \cdot \ell_P$
$L_0 = \xi \cdot \ell_P$	sub-Planck saturation	$2{,}16 \times 10^{-39}$	$n = +1{,}0$
ℓ_P	Planck length	$1{,}62 \times 10^{-35}$	$n = 0$
$\bar{\lambda}_e$	electron Compton wavelength	$3{,}86 \times 10^{-13}$	$n = -5{,}77$
L_ξ	Casimir vacuum scale	$1{,}00 \times 10^{-4}$	$n = -7{,}95$
c/H_0	Hubble length	$1{,}38 \times 10^{26}$	$n = -15{,}72$

Table 1: The ξ-power ladder spans from sub-Planck to cosmological scales. L_ξ sits on a regular ladder, not as an isolated free parameter.

A geometric-mean observation (Doc 166):

$$\sqrt{\ell_P \cdot \tfrac{c}{H_0}} = 47{,}3\,\mu\text{m}, \qquad L_\xi = 100{,}2\,\mu\text{m}, \qquad \frac{L_\xi}{\sqrt{\ell_P \cdot c/H_0}} = 2{,}12\,. \tag{6}$$

5 The Casimir-Hubble Bridge Relation

FFGFT also predicts an algebraic relation between the Casimir scale and the Hubble constant. Combining Eq. (3) with the FFGFT prediction for H_0 in Doc 165, $\hbar H_0/(m_e c^2) =$

$(\pi/2)\,\xi^{10}$, gives:

$$\boxed{L_\xi \cdot \frac{H_0}{c} = \frac{\pi}{2}\left(\frac{15}{\pi^2}\right)^{1/4} \frac{m_e c^2}{k_B T_{\mathsf{CMB}}}\,\xi^{41/4}} \tag{7}$$

The exponent decomposes as

$$\frac{41}{4} = \underbrace{10}_{\text{H}_0\text{ exponent (Doc 165)}} + \underbrace{\frac{1}{4}}_{\text{Casimir exponent}},$$

which is the same exponent appearing in Doc 026 as $E_H = E_0 \cdot \xi^{41/4}$. Numerically, both sides of Eq. (7) agree to $10^{-14}\%$. The relation contains no free parameters; all quantities are either physical constants or the single ξ.

In numerical form:

$$L_\xi \cdot \frac{H_0}{c} = 7{,}241 \times 10^{-31}\,, \qquad \frac{m_e c^2}{k_B T_{\mathsf{CMB}}} = 2{,}000 \times 10^{9}\,. \tag{8}$$

6 Comparison with Experimental Casimir Data

Reported experimental Casimir measurements of effective L_ξ values (from Doc 091):

Configuration	Effective L_ξ	Reference
parallel plates	228 nm	Dhital 2024
sphere-plate	$1{,}75\,\mu$m	Xu 2022
further reported	$18\,\mu$m	various

Table 2: Reported experimental L_ξ values vary across configurations (228 nm to 18 μm), reflecting geometric differences ($F \propto 1/L^4$ for parallel plates, $F \propto 1/L^3$ for sphere-plate) and experimental conditions. The FFGFT prediction $L_\xi \approx 100\,\mu$m is the configuration-independent vacuum-CMB anchor; geometry-specific values are expected to differ by known factors.

7 Summary of Predictions

1. $L_\xi = 100{,}24\,\mu$m *follows* from ρ_{CMB} and a single dimensionless $\xi = 4/30{,}000$. No free fit parameters.

2. Substitution into the modified Casimir density Eq. (4) reproduces the standard $\pi^2\hbar c/(240\,d^4)$ exactly. Mathematical consistency with all standard Casimir measurements at the operational level is preserved.

3. L_ξ sits on a five-step ξ-power ladder spanning Planck to Hubble scale; it is approximately $2{,}12\times$ the geometric mean $\sqrt{\ell_P \cdot c/H_0}$.

4. The Casimir scale and the Hubble constant satisfy the algebraic bridge Eq. (7), with combined ξ-exponent $41/4 = 10 + 1/4$.

5. The corresponding frequency $c/L_\xi \approx 3$ THz lies in the same regime as Casimir-scale phenomenology discussed elsewhere in the IPI correspondence.

Document references

Doc 091 *Casimir-CMB unified description.* Derivation of L_ξ and the modified Casimir formula; consistency with the standard $1/d^4$ scaling.

Doc 165 *The Hubble ratio.* Derivation of $\hbar H_0/(m_e c^2) = (\pi/2)\,\xi^{10}$, yielding $H_0 \approx 66{,}8$ km/s/Mpc within 0.9% of the Planck value.

Doc 166 *The Casimir scale as cosmic link.* The ξ-power ladder and the bridge relation $L_\xi \cdot H_0/c$; explanation of the exponent $41/4$.

Doc 026 *Geometric cosmology.* The exponent $41/4$ in $E_H = E_0 \cdot \xi^{41/4}$.

All documents available at `https://github.com/jpascher/T0-Time-Mass-Duality` and as Zenodo release v1.0.13: `https://zenodo.org/records/20041529`.